

Three questions for Active Inference in the Extremes

charlie derr · October 15, 2026 · Binghamton · draft opening

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Active inference

What should the agent do next?

Beliefs, preferences, information

PROGRAMME

Predictive coding

How should a correction learn?

Iterative error-inference credit

MEASURED TRADE-OFFS

Heavy-tailed distributions

Where do consequences fall?

Frequency, severity, concentration

MEASURED TESTBED EVIDENCE

Frozen transformer prior + adaptive residual cap: implemented testbed for a broader proposal.

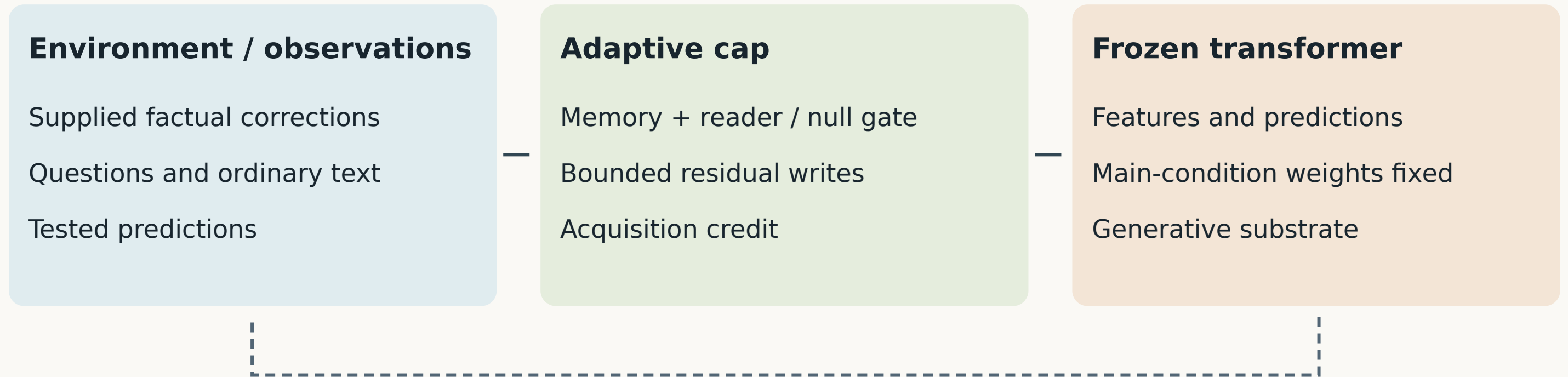
Submitted title retained in speaker source. A coupled agent and a heavy-tail family remain unestablished.

Claims: AI-programme, PC-v0, HT-readout

From active inference to our testbed

Implemented components; proposed policy loop

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PROPOSED: preferences + policy inference + information-seeking audits

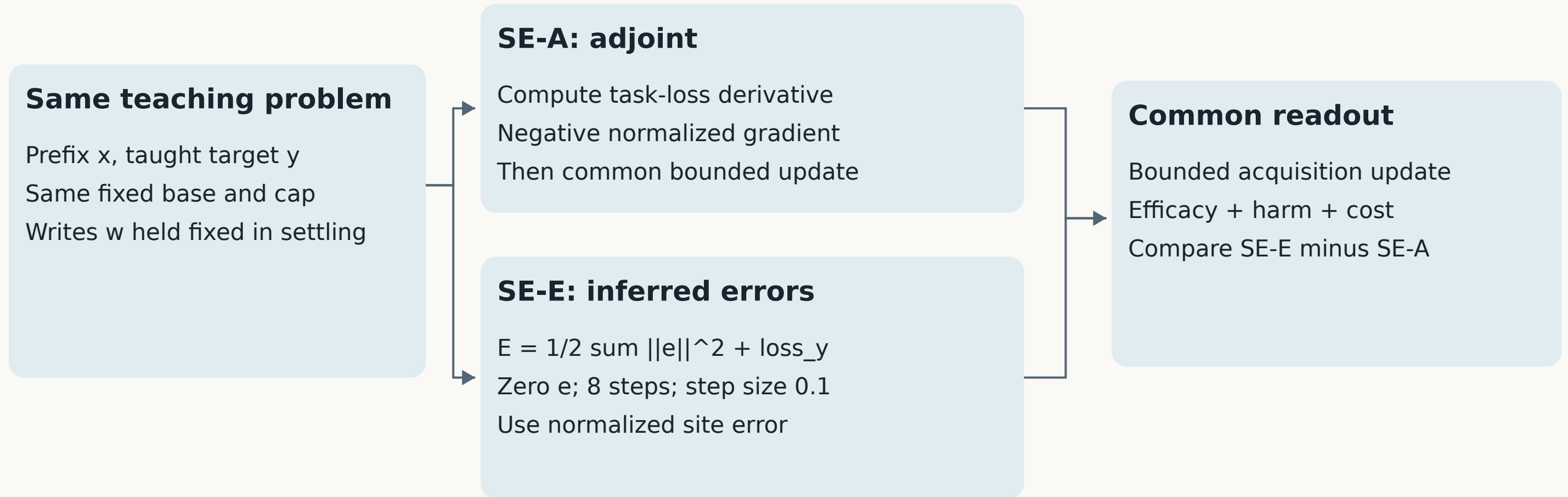
Interfaces are implemented. Coupled objective, blanket properties and policy loop remain proposed.

Claims: AI-loop, AI-testbed, AI-programme, AI-coupled-FE

Predictive coding supplies acquisition credit

Mechanism and acquisition comparisons measured; reader-training replication continues

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SD-24: penalize e before adding w; old write-site prior penalized e + w.

9 forward + 9 reverse evaluations per 8-step inference; JAX autodiff. Compare efficacy, harm, cost.

Claims: PC-mechanism, PC-SD24, PC-v0, PC-fixed-v5

A correction testbed with distinct controls

Three realizations × five dependent orders; endpoint depends on dataset

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Learned-reader condition

Frozen GPT-2 base

Fixed BP-trained reader

Adaptive bounded memory

Controlled comparisons

Random reader / stable cap

Live cap / matched update

S1 continued-base controls

Separate questions

Acquisition and retention

Locality and near misses

Ordinary-text probability harm

zsRE / CounterFact: 1,000 edits. MQuAKE: descriptive at 300. This reader study uses BP, not PC credit.

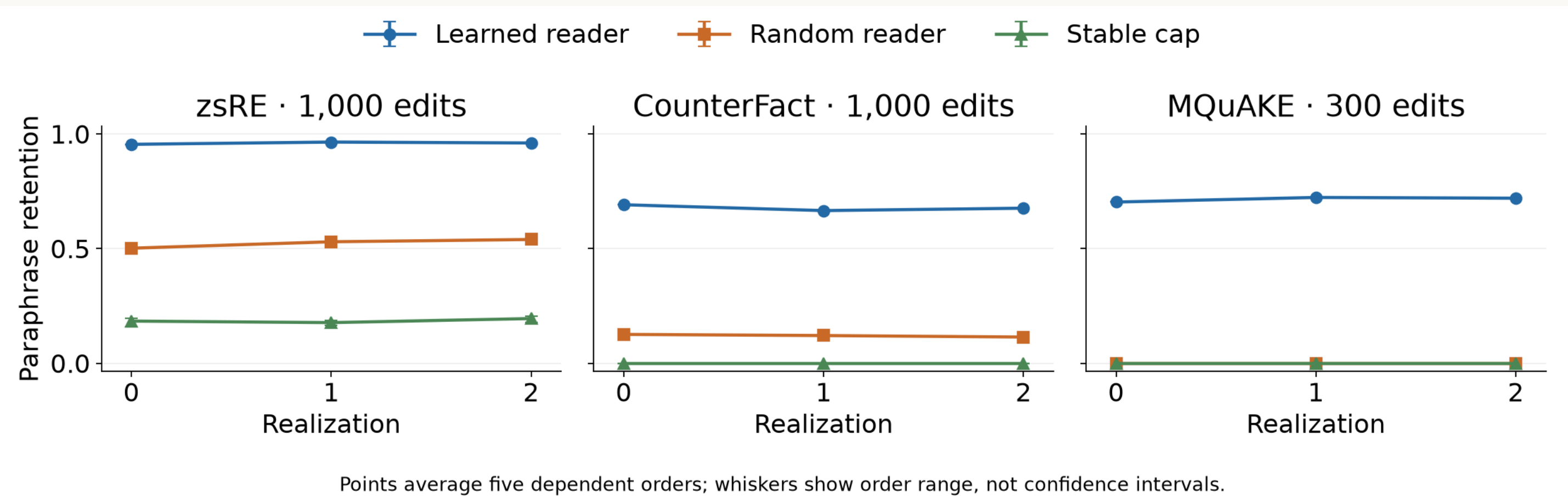
50 locality prompts do not exhaust specificity. Five orders within a realization are dependent.

Claims: R1-design, R1-controls, R1-fidelity

Retention improves, with limits to specificity

Completed triplet + 270-cell comparator snapshot; preliminary three-realization analysis

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Mean paraphrase retention: zsRE .9603; CounterFact .6780; MQuAKE .7156 (300 edits).

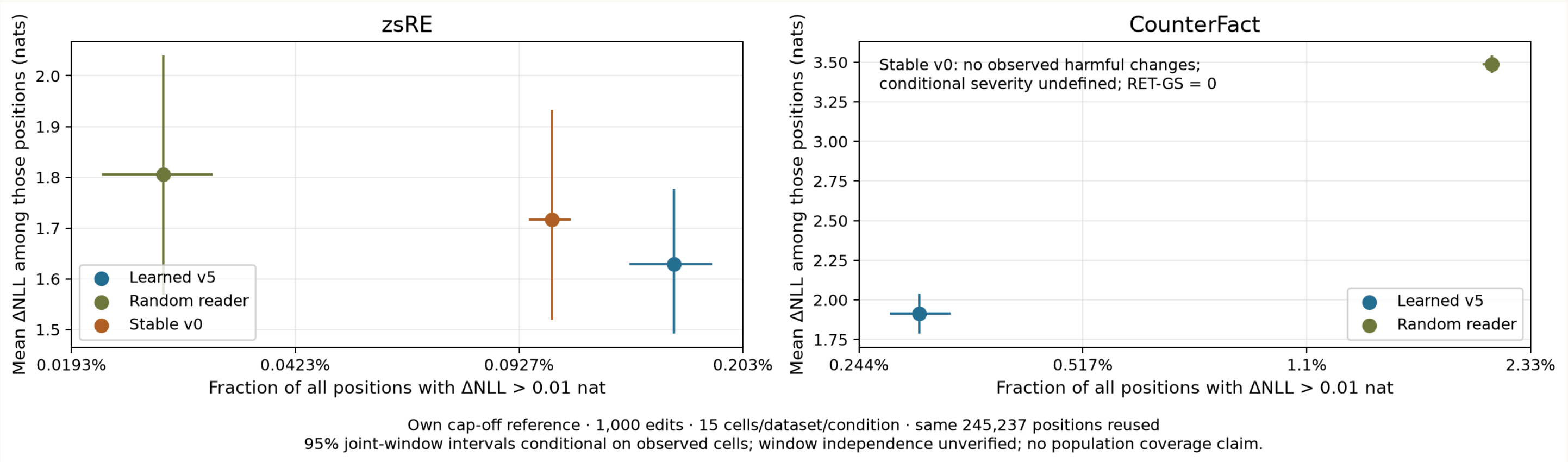
CounterFact vs stable: inconclusive. zsRE near-miss passes: 86 / 92 / 87 out of 100.

Claims: R1-triplet-summary, R1-specificity, R1-retention-zsre, R1-retention-counterfact

How often, and how severe?

HT-17: Stage-4 primary conditions; 1,000 edits; fixed ordinary-text inventory

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Frequency above 0.01 nat is not gate firing. Severity is conditional on exceeding that threshold.

Same positions reused; window intervals do not cover subject or seed uncertainty.

Claims: HT17-tails, HT-readout

What does observed tail shape add?

Exploratory finite-range fits; primary threshold 0.01 nat

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Learned v5

Illustrative 95% shape intervals

zsRE: -0.097 to 0.177

CF: -0.009 to 0.187

Stable v0 · zsRE

Illustrative interval: .390–.779

Held-out gain: .149–.594

nats/excess across 15 cells

Limits remain visible

Random CF: 10/15 failures

Mixture: all fits invalid

Sparse cells: no shape

Learned v5: little predictive gain over exponential. Stable v0: a more pronounced observed tail.

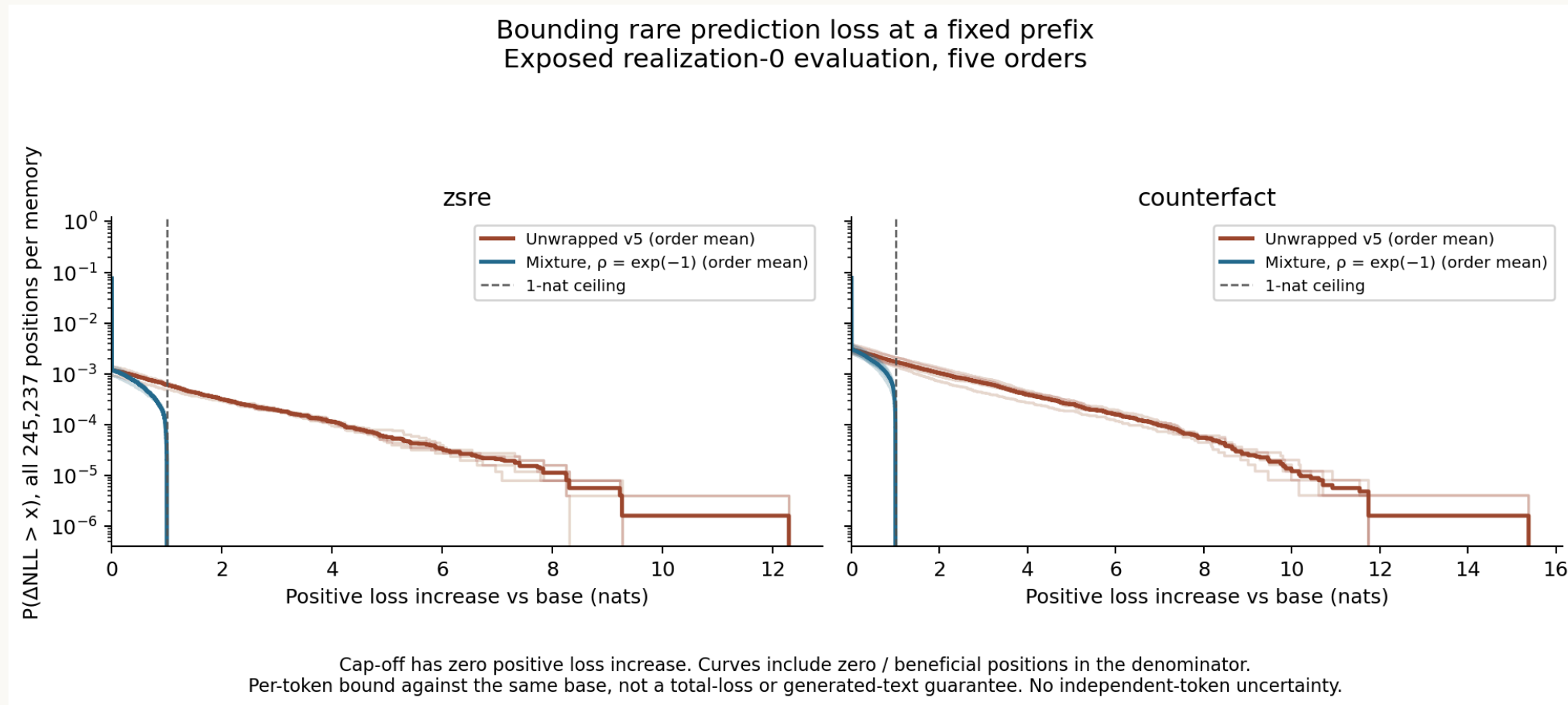
Conditional window intervals; illustrative r0/order100. No asymptotic class or infinite-variance finding.

Claims: HT17-tails

Two interventions against extreme prediction loss

κ : development success rule missed. Mixture: ten exposed memories pass.

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Mixture keeps retention within two points; CounterFact KL remains above 0.001.

κ pilot: bounded surprisal, not calibrated entropy. Mixture: analytic one-nat ceiling at shared prefix.

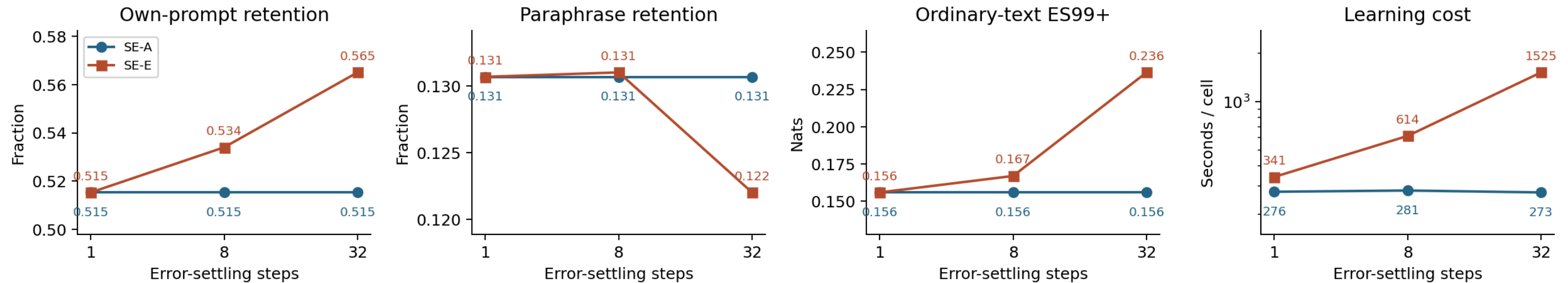
Claims: kappa-design, AW-B, HT17-tails

Predictive coding: retention, harm and learning cost

Depth controls: order100, three realizations per dataset; exposed historical streams.

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More settling: better taught-answer retention, higher cost and ES99+
zsRE · same order 100 · three realization means



Exposed S5. 1,000 edits; harm on 4,064 fixed positions. One-step endpoint scores equal adjoint; harm differs slightly numerically.
No confidence interval or depth selection. CounterFact is saturated / at a paraphrase floor; full results in the report.

Random: 3/993 immediate successes. Adjoint budget offered, underspent.

More own-prompt retention does not establish better generalization or equal-compute superiority.

Claims: PC-depth, PC-one-step, PC-random, PC-budget

Does PC credit transfer to the fixed v5 reader?

One exposed R1 realization per dataset; 300 edits; paired credit-rule intervention

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Paraphrase retention: Δ

zsRE: 0

CounterFact: -0.003333

100-edit values in notes

Positive-harm ES99: Δ

zsRE: -0.01397

CounterFact: 0.01123

Full 245,237-position readout

Acquisition + assay cost

SE-A stream s: 604.6

SE-E stream s: 776.7

Harm readout s: 5004

Same BP base, trained reader, gates, memory capacity and stream; fresh memory for each arm.

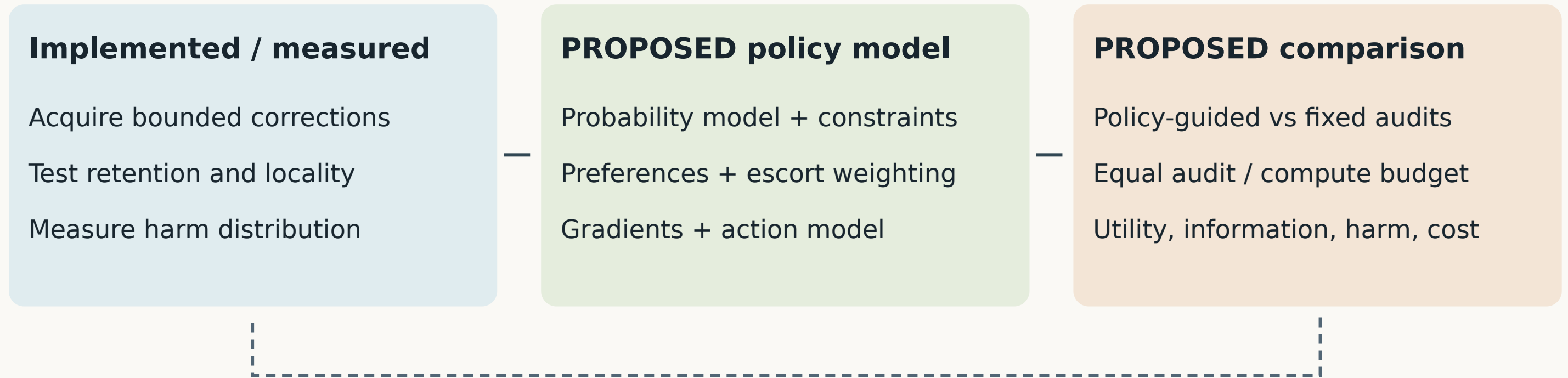
Separate reader training: 12/12 evaluations; 3 paired seeds; $\sim 100\times$ BP cost. CF paraphrase loss in all seeds.

Claims: PC-fixed-v5, PC-mechanism, HT-readout, PC-reader

What should the agent do next?

Proposed active-inference extension; no new pre-October-9 commitment

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Learning from observed failures motivates an experiment on autonomous information seeking.

Coupled free energy is a candidate objective. No implemented policy loop or blanket theorem.

Claims: AI-programme, AI-next, AI-coupled-FE, HT17-tails, PC-v0, PC-fixed-v5

What we learned, what remains, what it took

Measured credit and tail interventions; autonomous audit policies remain proposed.

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Active inference

Concrete correction testbed

Autonomous audit policy

REMAINS PROPOSED

Predictive coding

Credit: harm/cost trade-offs

Reader: 3 seeds; $\sim 100\times$ cost

CF PARAPHRASE LOSS: ALL SEEDS

Extremes

Mixture bounds token harm

All ten memories pass rule

MEASURED; FAMILY UNPROVEN

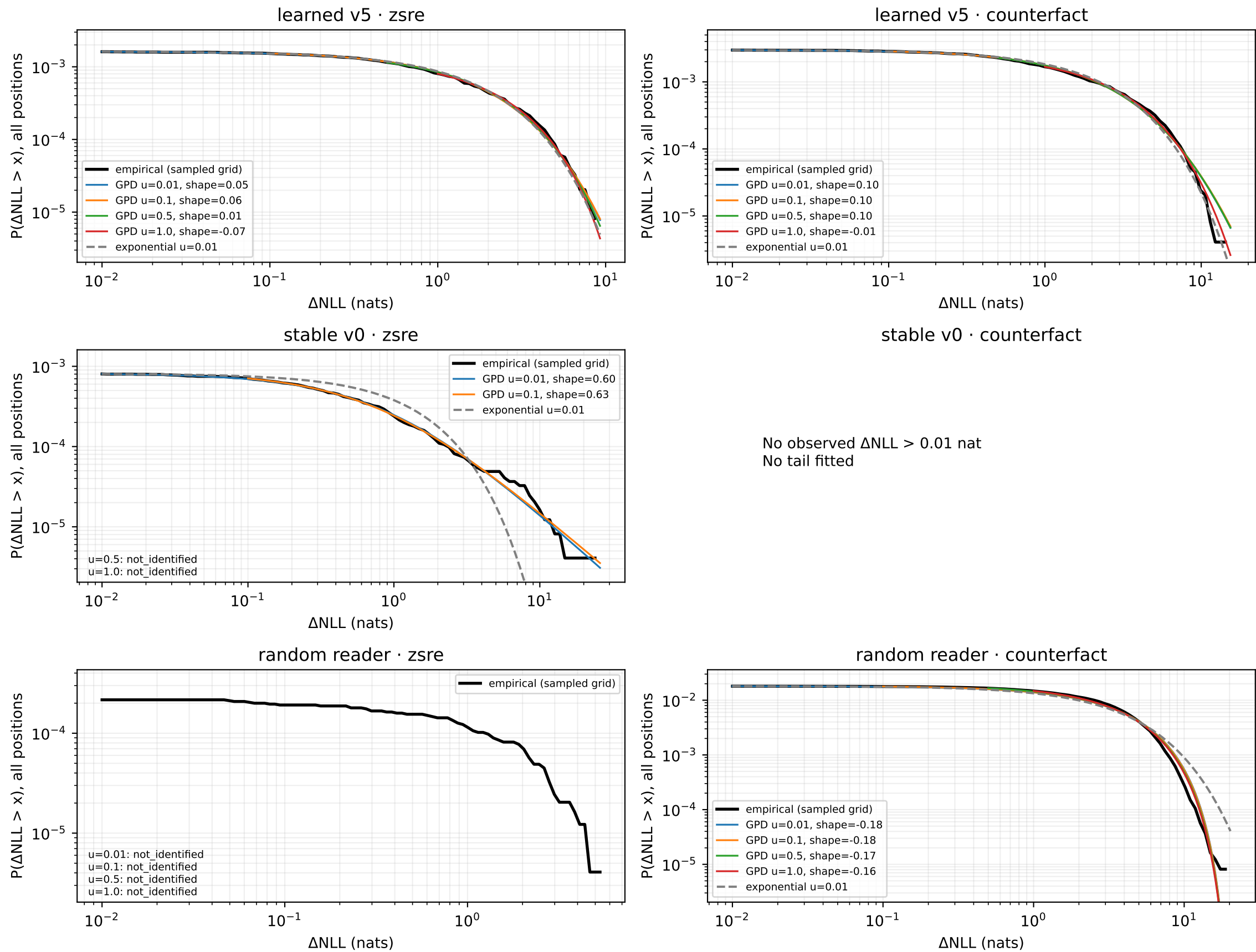
270-cell snapshot: 392.42 process-hours, not elapsed GPU hours. PC costs reported separately.

GPT-2 small (124M). GPU work complete Oct 4; review freeze Oct 9; talk Oct 15. Upper-layer results in backup.

Claims: AI-next, PC-depth, PC-budget, AW-B, HT13-corrected, talk-scope, model-scale, PC-reader, R-extension, AW-L

Finite-range survival and threshold sensitivity

Fixed illustration: realization 0, order 100; all cells retained in tables



Curves are fitted conditional excess models $\times$ observed exceedance probability. Invalid/sparse fits omitted explicitly; zero survival is not floored.
Curves do not identify an asymptotic law or complexity class; see held-out-window scores and realization/order variation in the report.